LINEAR PROGRAMMING: Graphical Method

Thirty-second Lecture by Prof. Uday S. Dixit, Department of Mechanical Engineering, IIT Guwahati



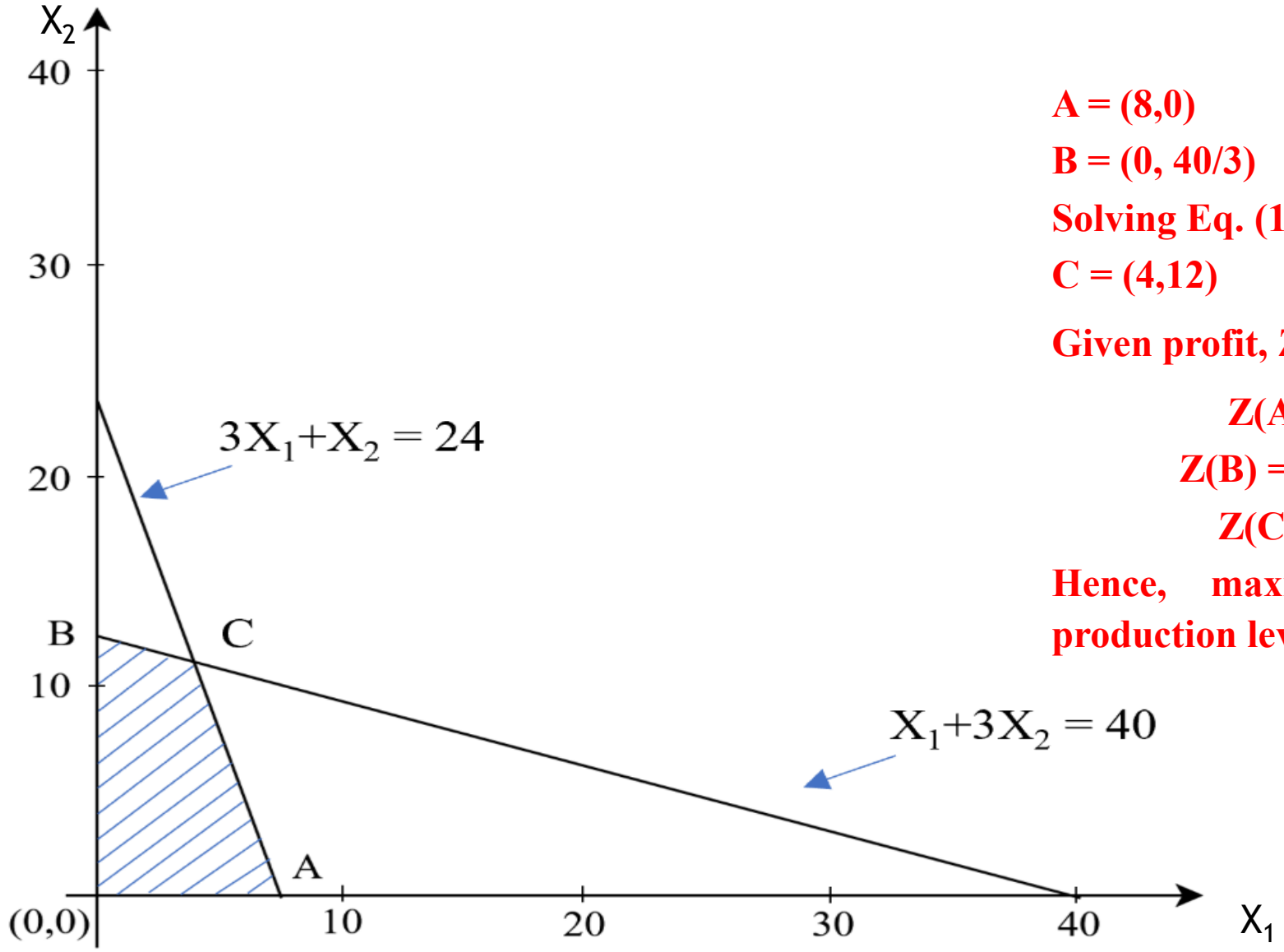
A manufacturer produces two types of products, 1 and 2, at production levels of X1 and X2. The profit is given as $2X_1+5X_2$. The production constraints are:

$$X_1+3X_2 \leq 40 \qquad (1)$$

$$3X_1+X_2 \leq 24 \qquad (2)$$

$$X_1 \geq 0,\ X_2 \geq 0,$$

Using graphical method, find out the production level in order to maximize profit? What is the maximum possible profit?



A = (8,0)

B = (0, 40/3)

Solving Eq. (1) and (2), we get

C = (4,12)

Given profit, $Z = 2X_1+5X_2$

$Z(A) = 2\times8+5\times0 = 16$

$Z(B) = 2\times0+5\times40/3 = 66.67$

$Z(C) = 2\times4+5\times12 = 68$

Hence, maximum profit is 68 at production levels of $X_1 = 4$ and $X_2 = 12$.

At the optimal solution $(X_1, X_2) = (4,12)$, we substitute in both constraints:

First Constraint:

$$X_1+3X_2 = 4+3\times12 = 40$$

Since the left-hand side equals the right-hand side, this constraint is active.

Second Constraint:

$$3X_1+X_2 = 3\times 4+12 = 24$$

Again, the left-hand side equals the right-hand side, this constraint is also active.

Non-negativity Constraints:

$$X_1 \geq 0, X_2 \geq 0$$

Since $X_1=4 > 0$ and $X_2 =12> 0$ at the optimal solution, these constraints do not limit the solution, so they are inactive.

To calculate unit worth of resource for constraint $X_1+3X_2 \leq 40$, we increase the RHS of constraint by 1 unit and solve the following LP:

Maximize $2X_1+5X_2$.

subject to

$X_1+3X_2 \leq 41$
$3X_1+X_2 \leq 24$
$X_1 \geq 0, X_2 \geq 0$

Solving we get $X_1 = 3.875$ and $X_2 = 12.375$. Maximum profit, $Z = 69.625$.

Unit worth of resource for constraint $X_1+3X_2 \leq 40$ = New profit – Original profit = 69.625 – 68 =1.625.

To calculate unit worth of resource for constraint $3X_1+X_2 \leq 24$, we increase the RHS of constraint by 1 unit and solve the following LP:

Maximize $2X_1+5X_2$.

subject to

$X_1+3X_2 \leq 40$
$3X_1+X_2 \leq 25$
$X_1 \geq 0, X_2 \geq 0$

Solving we get $X_1 = 4.375$ and $X_2 = 11.875$. Maximum profit, $Z = 68.125$.

Unit worth of resource for constraint $3X_1+X_2 \leq 24$ = New profit – Original profit = 68.125 – 68 = 0.125.

Two Graphical solution methods

- Extreme point Method

The extreme point method focuses on evaluating the objective function at the corner points (extreme points) of the feasible region.

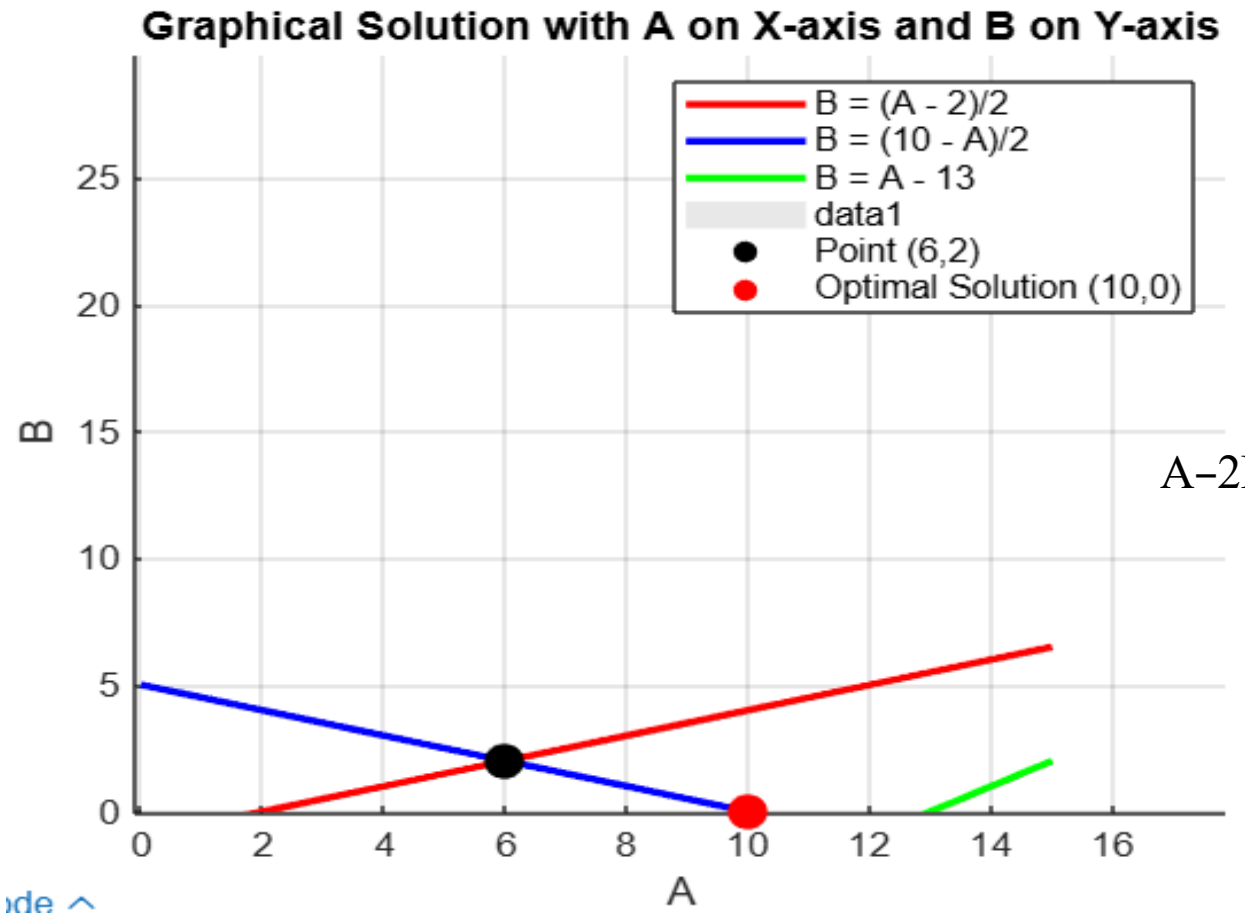
- Iso-Profit (Cost) Method

The iso-cost method involves drawing lines (iso-cost lines) representing the objective function and moving them to find the optimal solution.

Solve the following problem graphically:
Maximize $z=4A+2B$

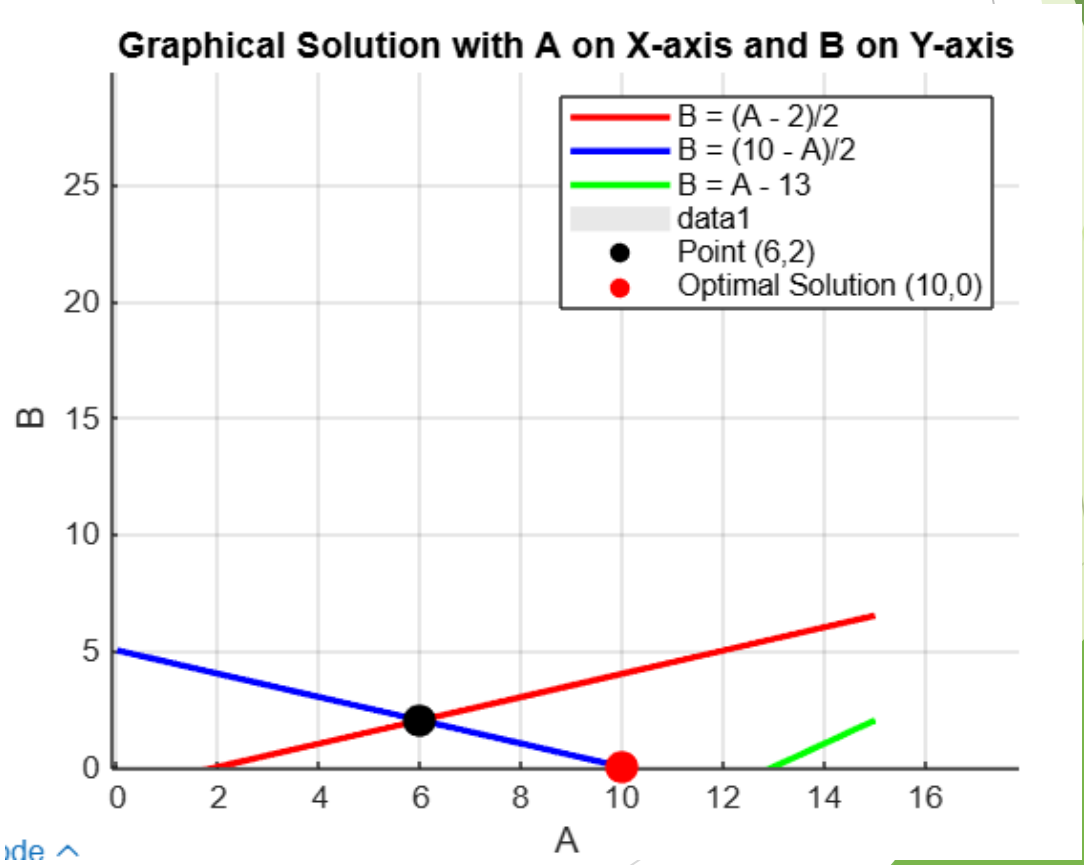
Subject to $A-2B\geq 2$; $A+2B=10$; $A-B\leq 13$; $A, B\geq 0$.

Suppose the objective function is changed $z=xA+yB$. Find out the range of x/y, in which same solution (i.e. same values of A and B) will be optimal.



$A-2B\geq 2;\ A+2B=10;\ A-B\leq 13;\ A, B\geq 0$

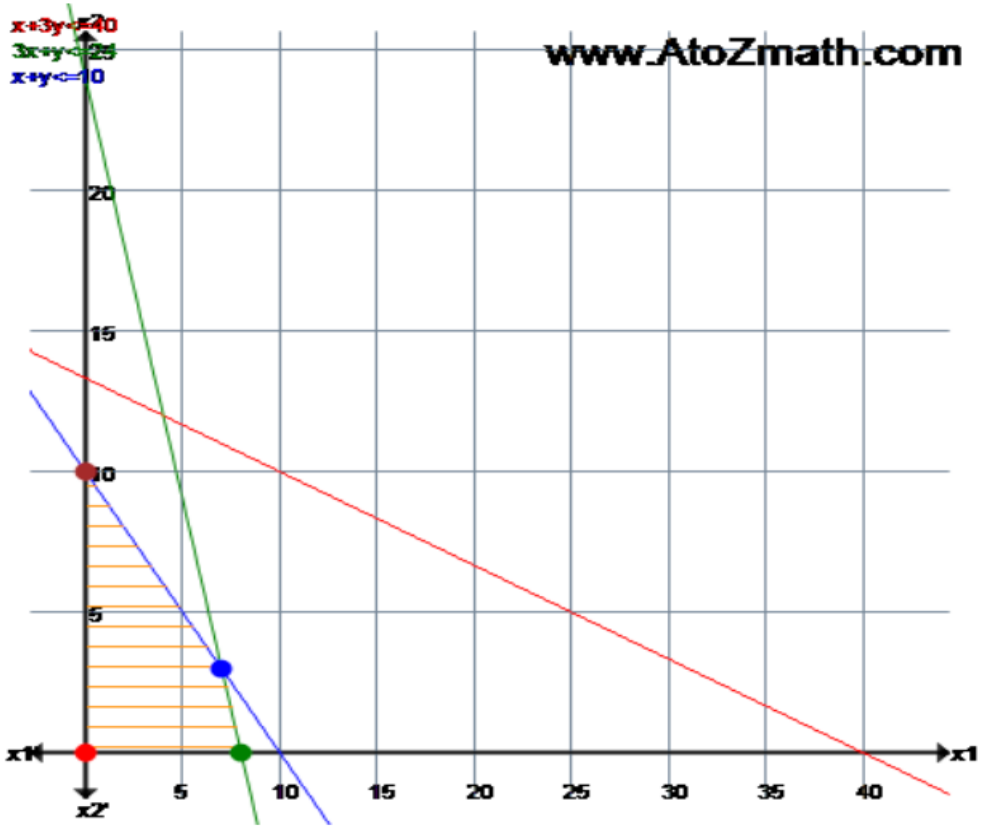
Blue line is A+2B=10. Solution has to lie on it. There are two corner points: Red point: A=10, 0; Black point: A=6, B=2. Function values on this point are 40 and 28, respectively. Hence, the optimum solution is 40 with A=10 and B=0. For general case, for first point (i.e A=10 and B=0) z=10x and for second point (i.e A=6 and B=2) z=6x+2y. As the first point is maximum, $10x \geq (6x+2y) \Rightarrow 4x \geq 2y \Rightarrow x \geq 0.5y \Rightarrow x/y \geq 1/2$. Range is 0.5 to infinity.



Maximize $2x_1 + 5x_2$

s.t.

$$x_1 + 3x_2 \le 40,$$
$$3x_1 + x_2 \le 24,$$
$$x1 + x2 \le 10,$$
$$x_1 \ge 0,\ x_2 \ge 0.$$



The value of the objective function at each of these extreme points is as follows:

Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $z = 2x_1 + 5x_2$
$O(0, 0)$	$4 \rightarrow x_1 \geq 0$ $5 \rightarrow x_2 \geq 0$	$2(0) + 5(0) = 0$
$A(8, 0)$	$2 \rightarrow 3x_1 + x_2 \leq 24$ $5 \rightarrow x_2 \geq 0$	$2(8) + 5(0) = 16$
$B(7, 3)$	$2 \rightarrow 3x_1 + x_2 \leq 24$ $3 \rightarrow x_1 + x_2 \leq 10$	$2(7) + 5(3) = 29$
$C(0, 10)$	$3 \rightarrow x_1 + x_2 \leq 10$ $4 \rightarrow x_1 \geq 0$	$2(0) + 5(10) = 50$

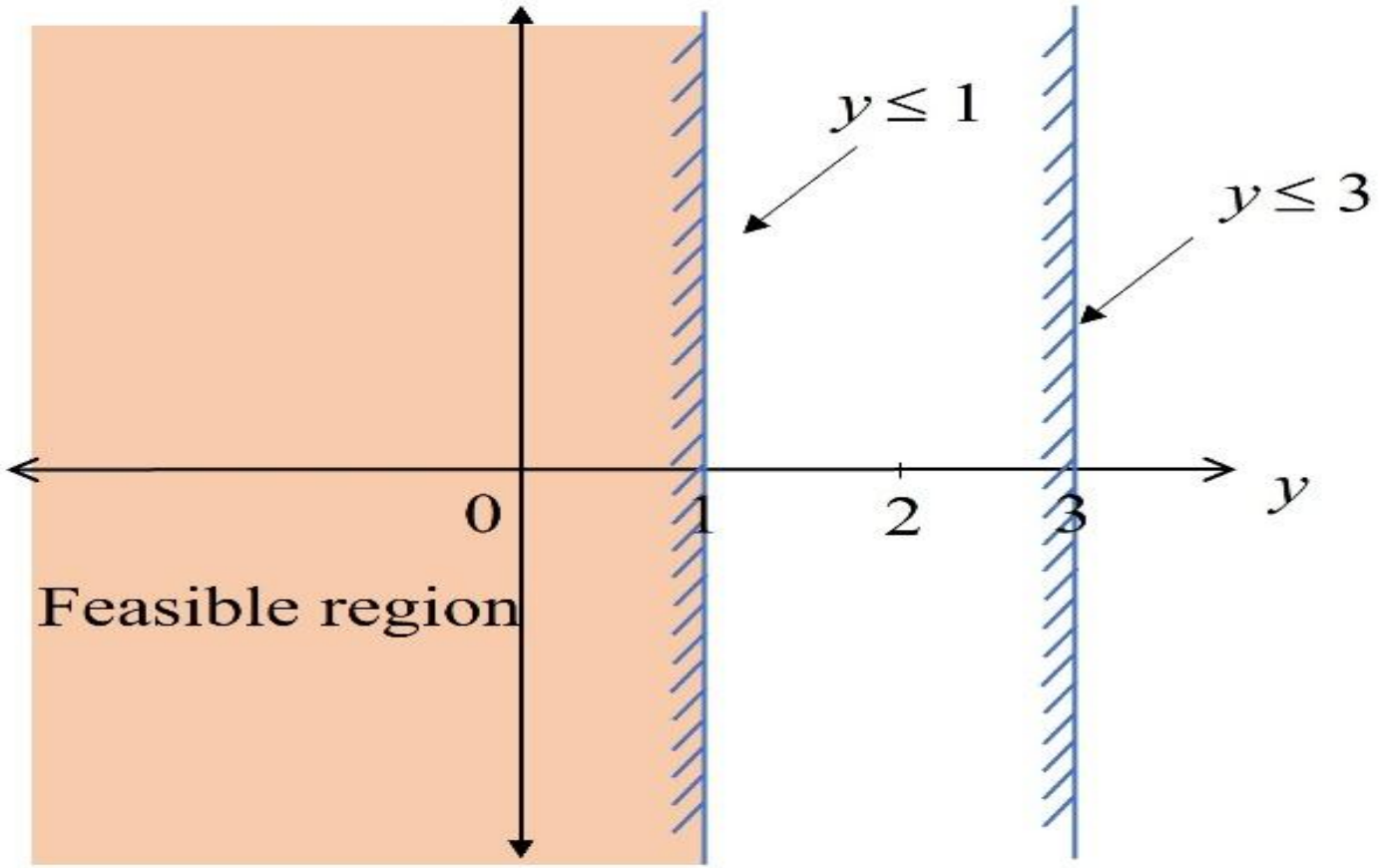
Maximize $w = 4y$

Subject to

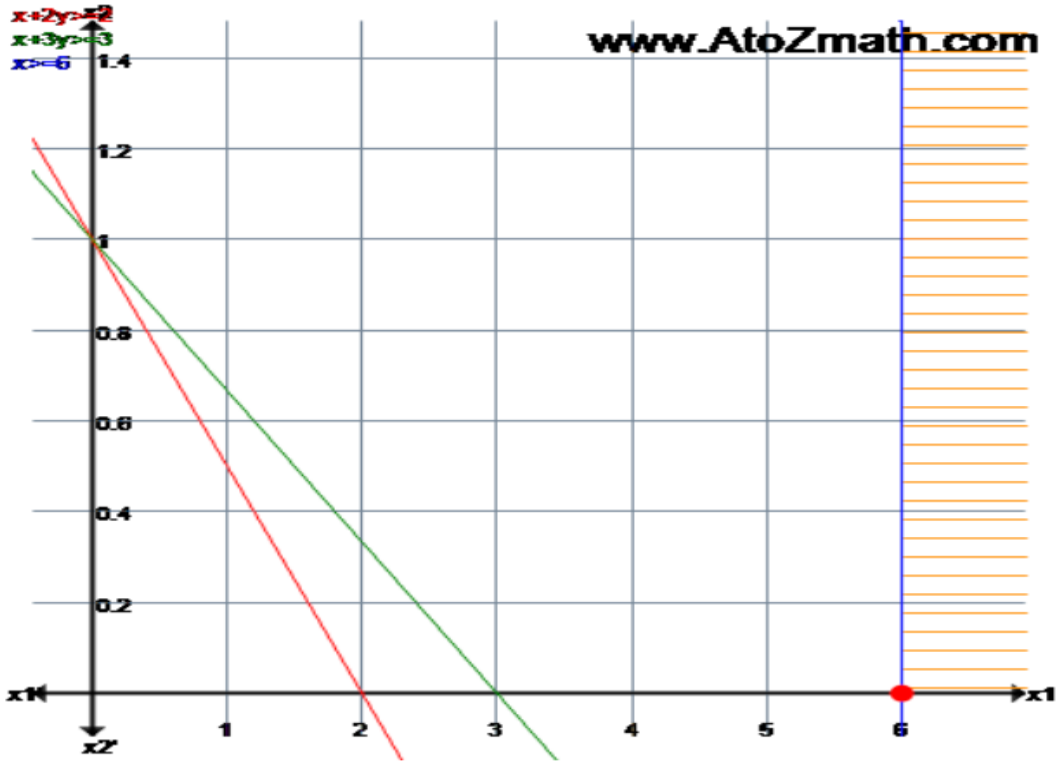
$y \leq 1$,

$y \leq 3$,

y is unrestricted.



Minimize $w = y_1 + y_2$ Subject to $y_1 + 2y_2 \geq 2; y_1 + 3y_2 \geq 3; y_1 \geq 6; y_1, y_2 \geq 0$



The value of the objective function at each of these extreme points is as follows:

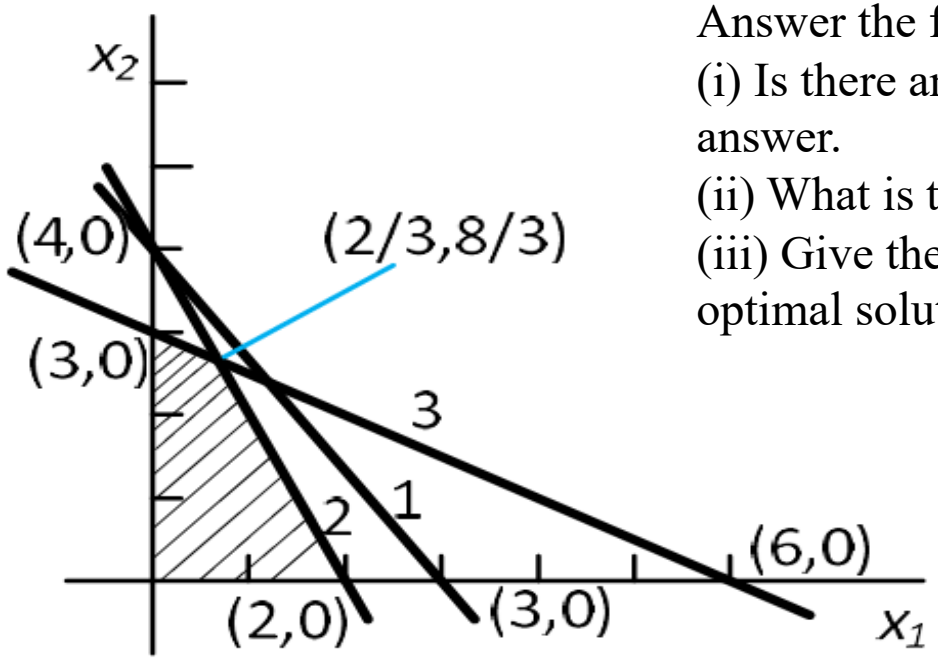
Extreme Point Coordinates (x_1, x_2)	Lines through Extreme Point	Objective function value $Z = x_1 + x_2$
$A(6, 0)$	$3 \rightarrow x_1 \geq 6$ $5 \rightarrow x_2 \geq 0$	$1(6) + 1(0) = 6$

The miniimum value of the objective function $Z = 6$ occurs at the extreme point $(6, 0)$.

Hence, the optimal solution to the given LP problem is : $x_1 = 6, x_2 = 0$ and min $Z = 6$.

The graphical solution of a two-variable linear programming problem is depicted in the figure. Both the variables are in the first quadrant. The linear constraints are labelled as 1, 2 and 3, and the feasible region has been hatched. The objective is

$$\text{Maximize } z = 2x_1 + x_2$$



Answer the following questions:

(i) Is there any redundant constraint in this problem? Justify your answer.

(ii) What is the value of the optimal solution?

(iii) Give the coordinates of all the points at which you can get the optimal solution.

$$\left.\begin{aligned} \hat{x}_1 &= \alpha(2/3) + (1-\alpha)(2) = 2 - \frac{4}{3}\alpha \\ \hat{x}_2 &= \alpha\left(\frac{8}{3}\right) + (1-\alpha)(0) = \frac{8}{3}\alpha \end{aligned}\right\}, 0 \le \alpha \le 1$$

https://web.mit.edu/15.053/www/AppliedMathematicalProgramming.pdf

